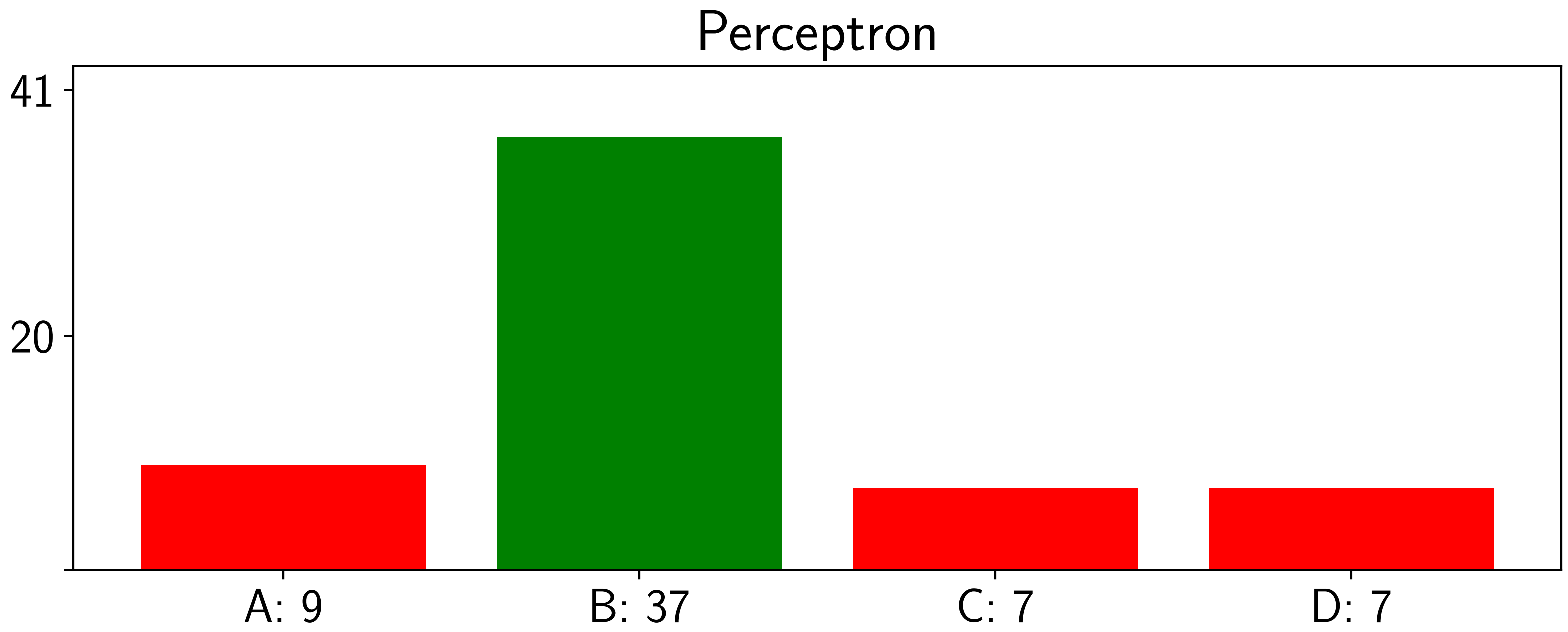


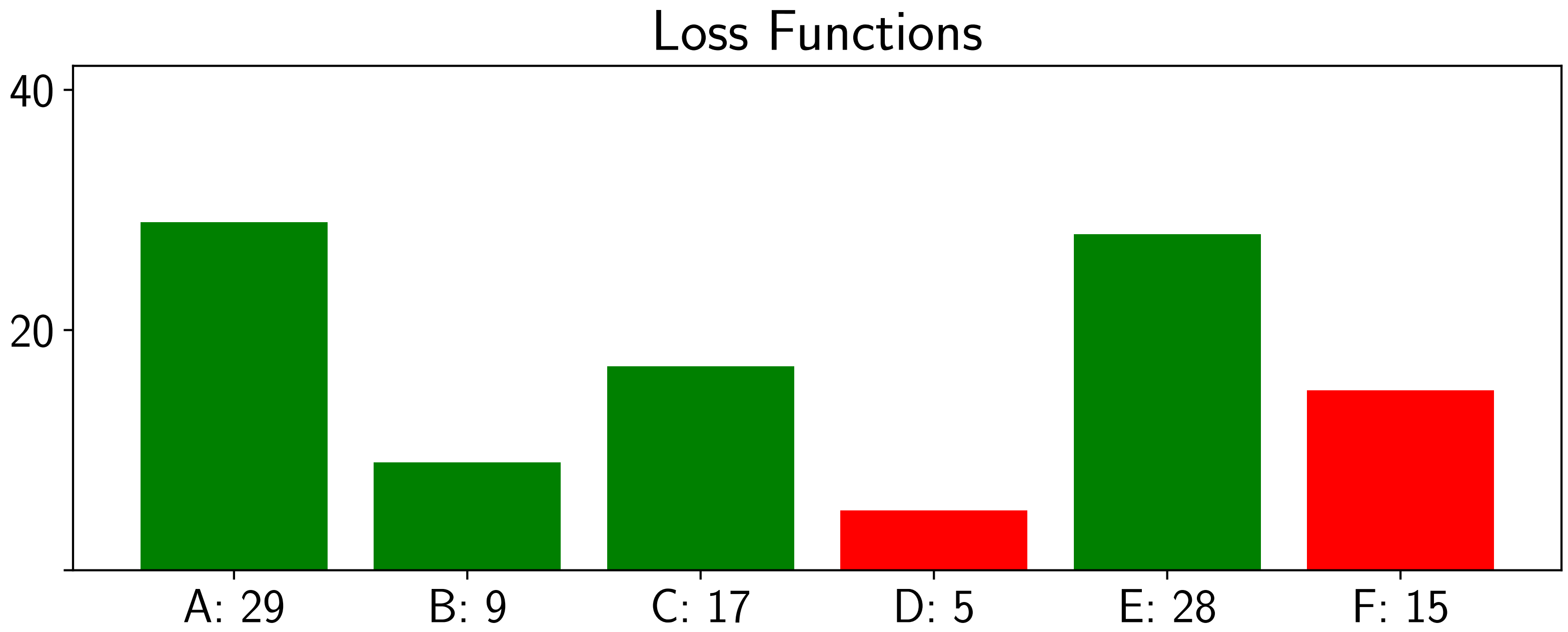
Please mark all correct answers.

A The perceptron only works on binary input.

B The perceptron produces binary output.

C The perceptron learning will converge when the data is not linearly separable.

D The perceptron learning will provide the optimal solution.



Please mark all correct answers.

A The goal of machine learning is to reach the minimum of the loss function.

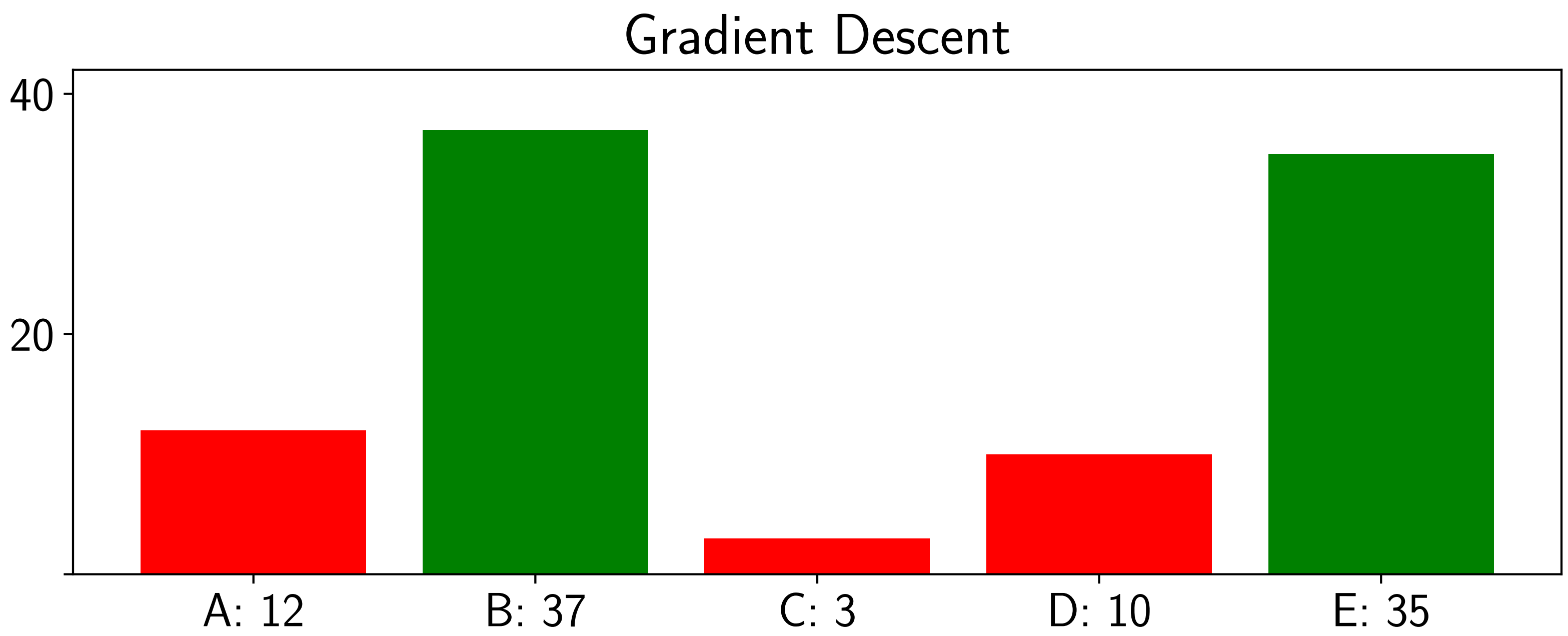
B The perceptron loss function has a single minimum.

C For separable data, this minimum is always reached.

D The minimum of the perceptron loss is reached by a single set of weights.

E For a linear regression model, there exists exactly one minimum of the loss function.

F Gradient descent is required to find the minimum of the loss function in a linear regression task.



Please mark all correct answers.

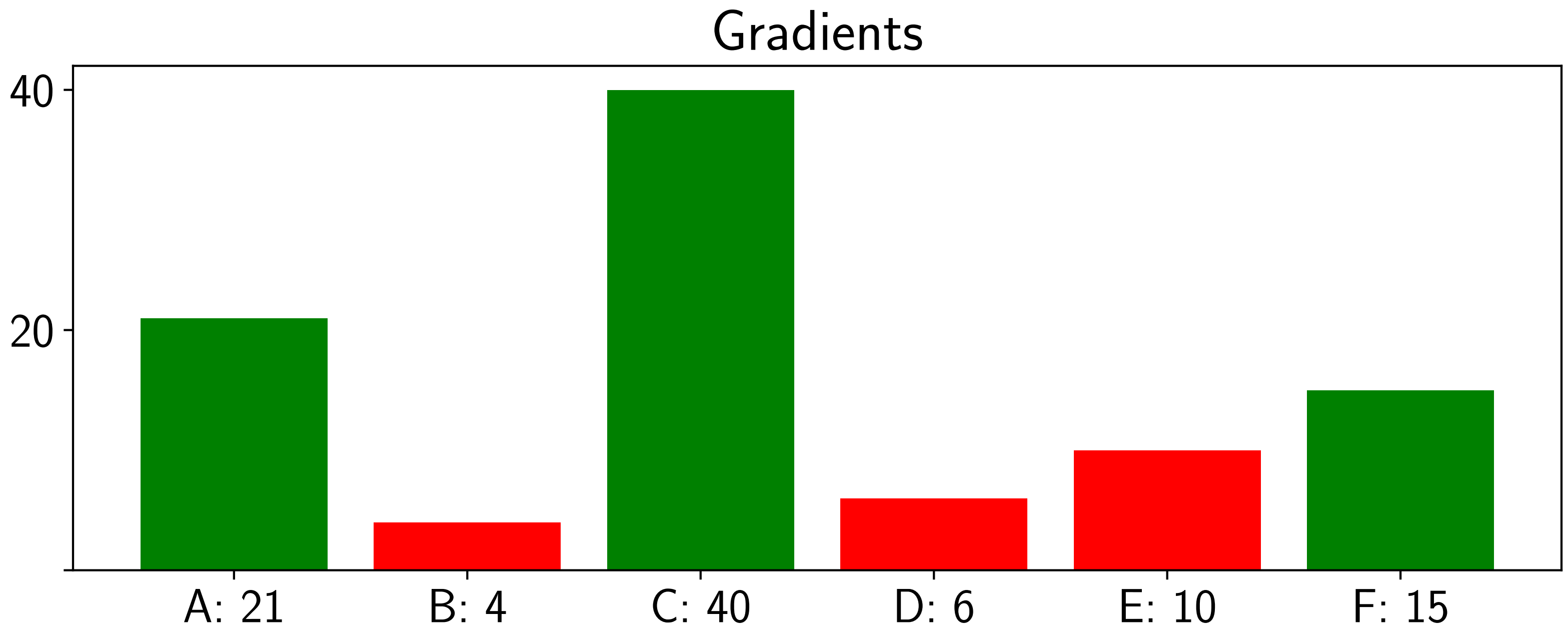
A The gradient is pointing to the steepest decrease in the loss function.

B The gradient contains the partial derivatives of the loss with respect to all learnable parameters.

C The quickest processing of gradient descent can be achieved with a large learning rate η.

D Gradient descent always finds a local minimum.

E An adaptive learning rate can be used to speed up convergence.



Which derivative rules are always applied for computing the gradient via error backpropagation?

A Sum rule

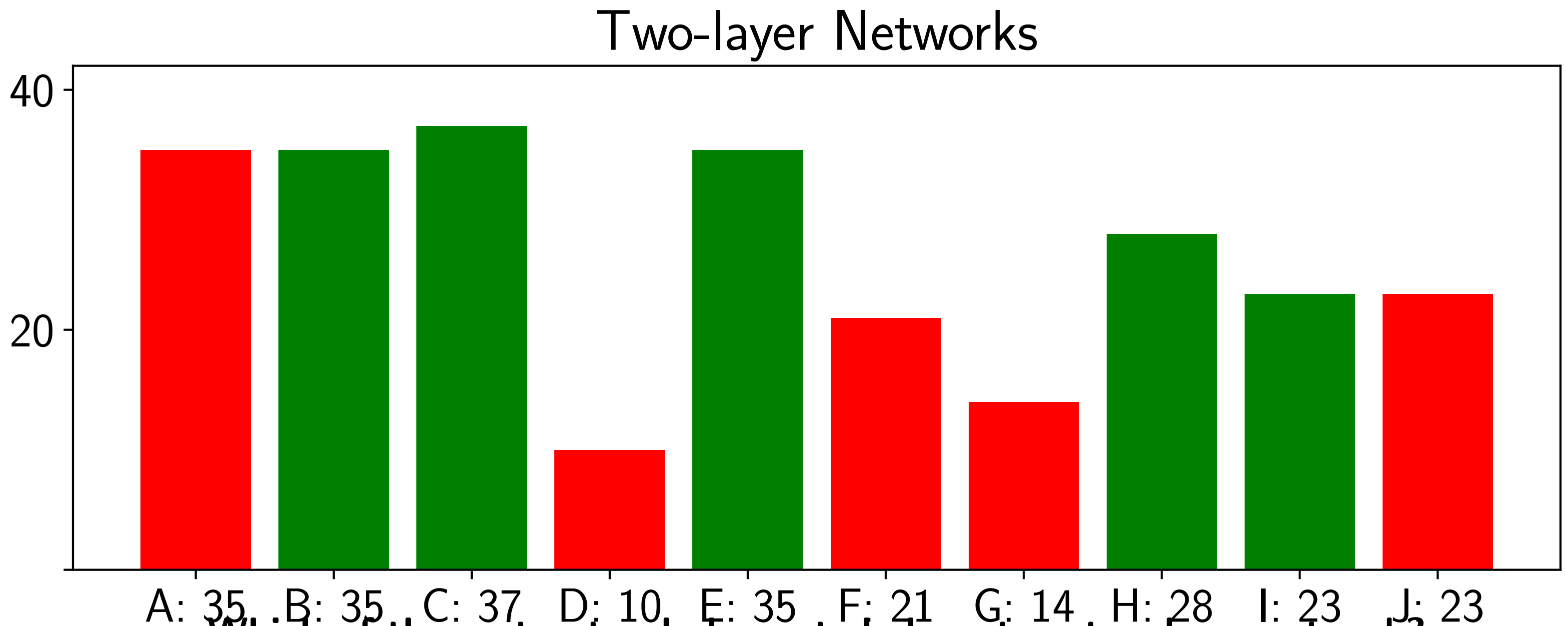
B Exponential rule

C Chain rule

D Square rule

E Reciprocal rule

F Constant rule



Which of these structural elements belong to a two-layer network?

A Input neurons

B Layer weights

C Activation function

D Network gradient

E Hidden neurons

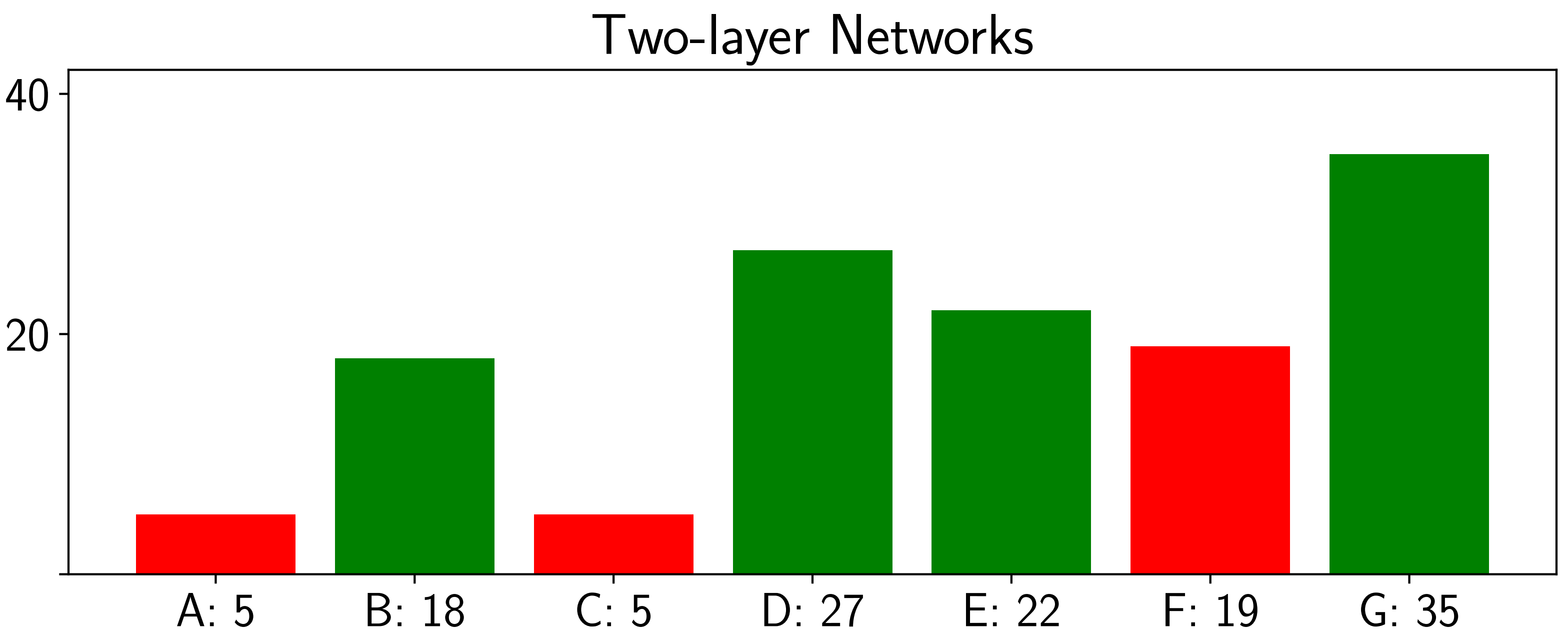
F Loss function

G Target values

H Output neuron(s)

I Bias neuron(s)

J Network output



Please mark all correct answers.

A A one-layer network with sigmoid activation can learn functions of arbitrary shape.

B A two-layer network with linear activation of the hidden neurons can only learn linear functions.

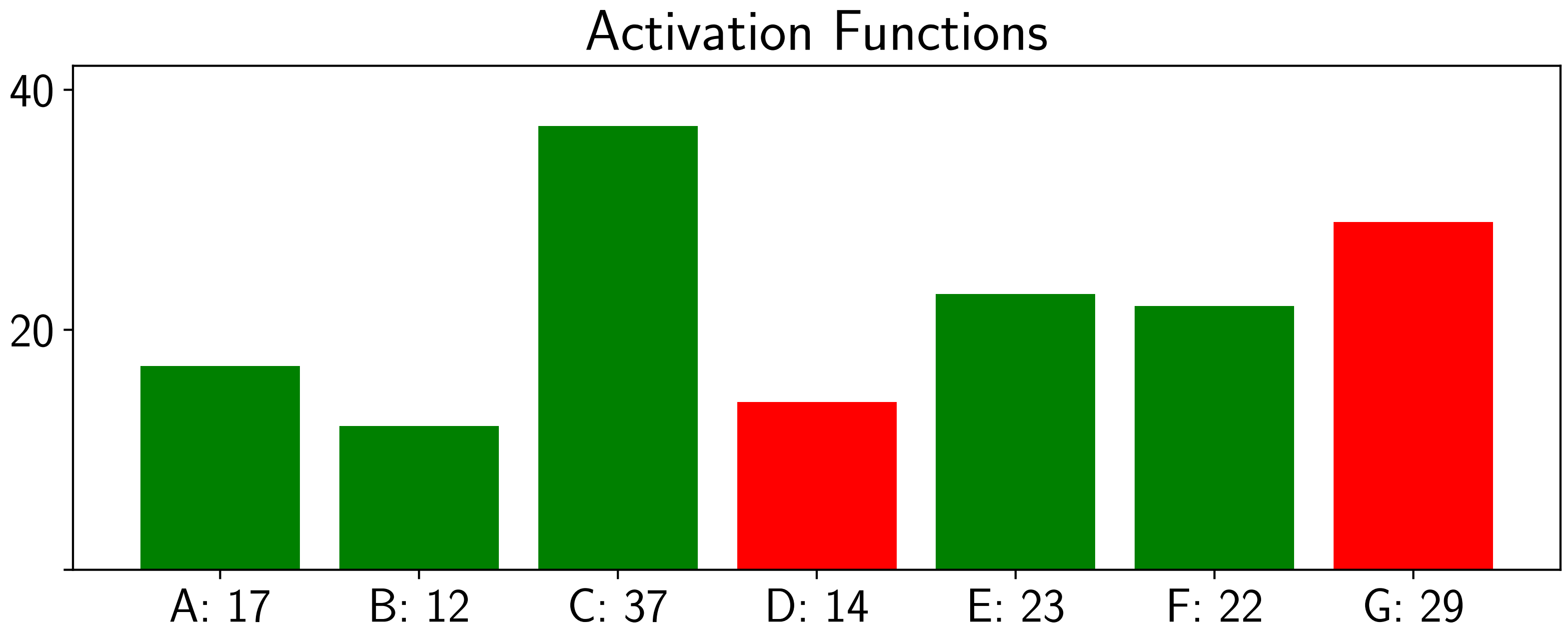
C A two-layer network with sigmoid activation function can only learn sigmoid-like functions.

D The number of hidden units determines the shape of the functions that can be learned by the network.

E The derivative of the sigmoid and the tanh activation functions can be computed from their respective outputs.

F The xor problem can be solved with a two-layer network that contains a single hidden neuron.

G With sufficiently many hidden neurons, any function can be approximated to any reasonable accuracy.



Which of these functions can work as activation functions?

A Linear Function $g(x) = x$

B Absolute Function: $g(x) = |x|$

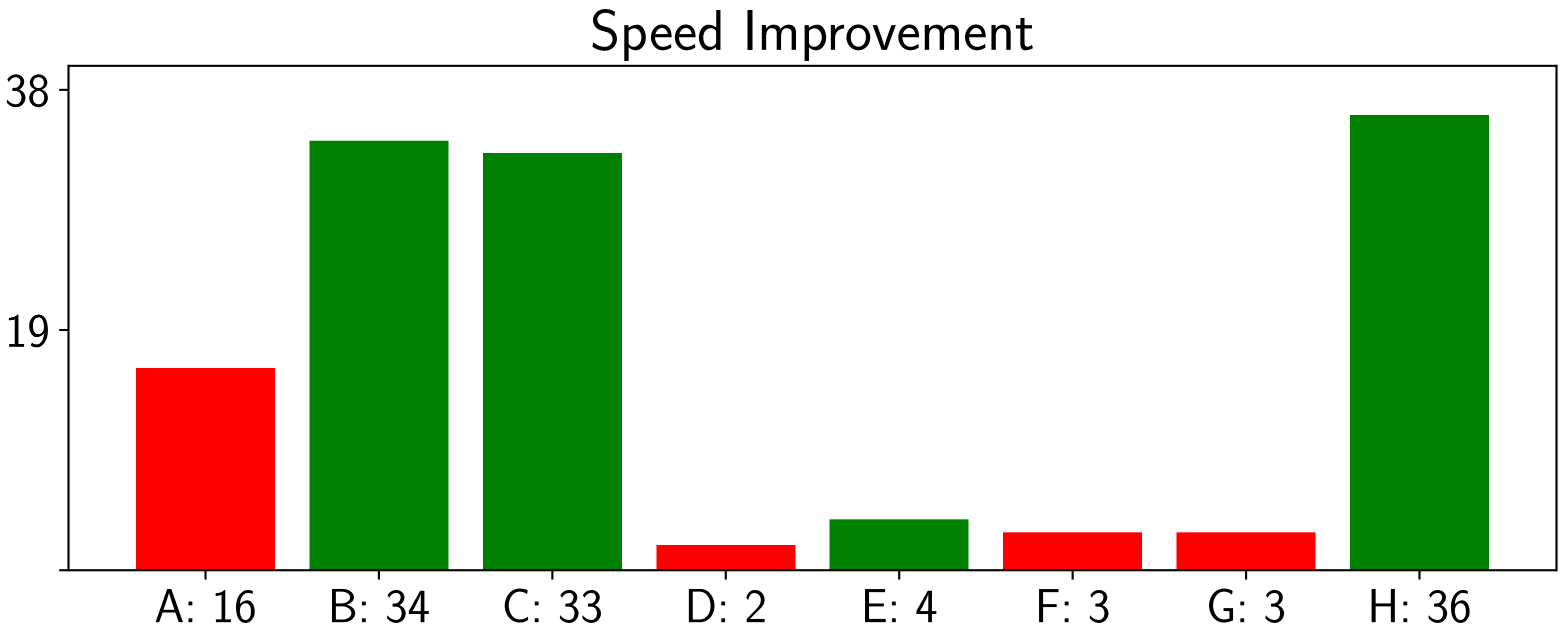
C Logistic Function: $g(x) = \frac{1}{1+e^{-x}}$

D Reciprocal function: $g(x) = \frac{1}{x}$

E Sine Function: $g(x) = \sin(x)$

F Gaussian Function: $g(x) = e^{-x^2}$

G Tangent function: $g(x) = \tan(x)$



How can we improve the training speed for one iteration over the dataset?

A Increase the learning rate

B Use matrix multiplications

C Avoid data reallocation

D Increase the number of hidden neurons

E Increase the first layer weight matrix to produce $K+1$ outputs

F Compute the sum over all samples

G Use a linear activation function

H Use stochastic gradient descent